

FacilityOps AI Platform

Predictive Maintenance & Role-Based Facility Management System

Domain: IIoT, Predictive Maintenance & Facility Management

Architecture: RBAC + Streamlit Multi-Page Web Application

1. Executive Summary

FacilityOps AI Platform is a predictive maintenance and facility operations web application designed to monitor industrial machine health, identify potential component failures, automate preventive maintenance scheduling, manage technician work orders, and use a local Artificial Intelligence model through Ollama for maintenance diagnostics and recommendations.

The platform analyzes air temperature, process temperature, rotational speed, torque, and tool wear to support a shift from reactive maintenance to data-driven proactive maintenance.

2. Problem Statement

Industrial equipment can experience unexpected failures because of excessive tool wear, abnormal temperature conditions, unsuitable power ranges, or mechanical overstrain. FacilityOps brings monitoring, analysis, work-order tracking, preventive scheduling, authentication, and local AI assistance into one application.

3. Project Objectives

- Monitor important machine operating parameters.
- Identify and analyze machine failure modes.
- Provide dashboards for operational and analytical understanding.
- Separate administrative and technician responsibilities using RBAC.
- Manage maintenance work orders from creation through completion.
- Organize preventive maintenance schedules and inspection checklists.
- Use Ollama-based local AI for machine diagnostic summaries and recommendations.
- Generate PDF work orders for maintenance documentation.

4. Authentication & Role-Based Access Control

The application uses a unified Authentication Gateway with role-based access control. Passwords are protected using SHA-256 hashing, while user information is persisted through `data/users.json`.

Updated Evaluation Login Flow

The previous Demo Login / one-click login instruction has been removed.

- Launch the application and wait for the Authentication Gateway.
- For the Administrator, open Create Account / Sign Up first.
- Create the Administrator account and complete the required account details.
- Return to Sign In and log in using the newly created Administrator credentials.
- For the Technician, use the Technician account configured for the project and log in through the same Sign In gateway.

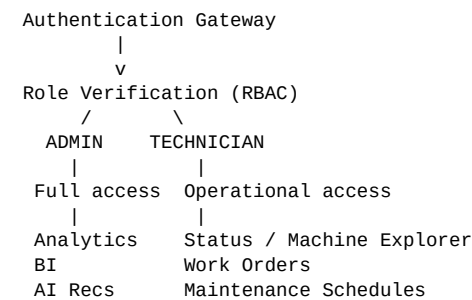
- The application checks the user's role and displays only the pages authorized for that role.

Role Comparison

Capability	Administrator	Technician
Home / Platform Overview	Granted	Granted
Maintenance Status	Granted	Granted
EDA	Granted	Restricted
Executive BI Dashboard	Granted	Restricted
Machine Explorer & AI	Granted	Granted
Work Orders	Granted	Granted
Maintenance Schedules	Granted	Granted
AI Recommendations	Granted	Restricted

5. System Architecture

The system is organized into presentation, data/analytics, visualization, AI, documentation/output, and security layers.



6. Dataset Architecture — AI4I 2020

The project processes the synthetic AI4I 2020 Predictive Maintenance Dataset containing 10,000 machine observations. Product quality variants are L (Low, 60%), M (Medium, 30%), and H (High, 10%).

Feature	Purpose
Air Temperature [K]	Ambient operating temperature
Process Temperature [K]	Internal machine temperature
Rotational Speed [rpm]	Spindle rotation rate
Torque [Nm]	Mechanical torque
Tool Wear [min]	Cumulative tool usage
Failure Modes	Binary failure targets

Failure Mode Logic

- **TWF — Tool Wear Failure:** tool wear approximately 200–240 minutes.
- **HDF — Heat Dissipation Failure:** temperature difference below 8.6 K and speed below 1380 rpm.
- **PWF — Power Failure:** Torque × Speed below 3500 W or above 90000 W.
- **OSF — Overstrain Failure:** Torque × Tool Wear exceeds the product/material threshold.
- **RNF — Random Failure:** documented 0.1% baseline random hardware failure rate.

7. Functional Modules

Authentication Gateway

components/auth_ui.py & utils/auth.py
Sign In, account creation, recovery, security questions, session management and RBAC.

Maintenance Status Dashboard

1_Maintenance_Status_Dashboard.py

Overdue tasks, upcoming 7-day schedules, open work orders and maintenance history.

EDA

2_EDA.py

Administrator-only statistical profiling, missing-value inspection, distributions, correlations and failure breakdowns.

Executive BI Dashboard

3_Dashboard.py

Administrator-only interactive Plotly views of operating ranges, thermal margins and failure clustering.

Machine Explorer & AI Inspector

4_Machine_Explorer.py

Equipment search, sensor gauges, inspection, Ollama diagnostics and PDF Work Orders.

Work Orders

5_Work_Orders.py

Kanban tracking across Open, In Progress and Completed, with repair actions and PDF export.

Maintenance Schedules

6_Maintenance_Schedules.py

Daily, weekly, monthly and operating-hours schedules with inspection checklists.

AI Maintenance Recommendations

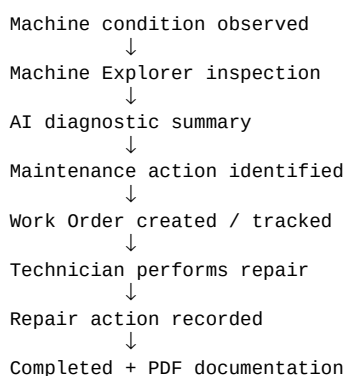
7_AI_Maintenance_Recommendations.py

Administrator-only AI briefing based on equipment wear and open work orders.

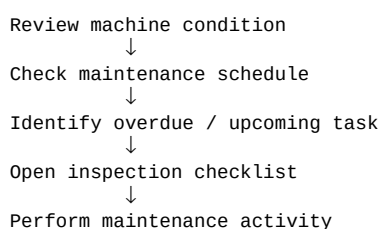
8. AI Integration — Ollama

The platform integrates a local LLM through Ollama for machine diagnostic summaries and maintenance recommendations. The documented local model integration includes llama3 / mistral.

9. Work Order Lifecycle



10. Preventive Maintenance Workflow



11. Security Design

- **SHA-256:** password hashing is used by the authentication layer.
- **User Persistence:** user information is stored through data/users.json.
- **Session Guards:** unauthorized page access is blocked.
- **RBAC:** page access is determined by the authenticated user's role.

12. Technology Stack

Technology	Purpose
Python 3.10+	Core programming language
Streamlit 1.59.2	Web application / dashboard framework
Pandas / NumPy	Data and numerical processing
Plotly Express / Graph Objects	Interactive visualization
Ollama	Local AI / LLM integration
llama3 / mistral	Documented local LLM models
ReportLab	PDF generation
JSON	User persistence
SHA-256	Password hashing

13. Installation & Launch

Prerequisite: Python 3.10 or higher.

Command:

```
streamlit run app.py
```

Evaluation: Create the Administrator account first. Then use normal Sign In for both Administrator and Technician accounts. Verify that restricted pages are hidden/blocked according to RBAC.

14. Suggested Evaluation Demonstration

- Authentication: create Administrator account and demonstrate normal login.
- Administrator: show Home, Maintenance Status, EDA and Executive BI Dashboard.
- Machine inspection: demonstrate machine search, sensor parameters and Ollama diagnostic summary.
- Maintenance execution: demonstrate Work Orders moving through Open → In Progress → Completed and PDF export.
- Preventive maintenance: demonstrate schedule frequency and inspection checklist.
- Technician: sign in and verify that operational pages remain available while Administrator-only analytics/recommendation pages are restricted.

15. Expected Project Outcomes

- Working Streamlit web application with role-based authentication.
- Predictive-maintenance data analysis and interactive dashboards.
- AI-assisted machine diagnostics.
- Maintenance work-order lifecycle tracking.
- Preventive maintenance scheduling.

- PDF maintenance documentation.
- Clear separation of administrator and technician workflows.
- End-to-end flow from machine data to maintenance action.

16. Advantages

- Centralized facility maintenance workflow
- Data-driven machine monitoring
- Role-specific access
- Local AI assistance
- Interactive dashboards
- Preventive maintenance planning
- Work-order lifecycle tracking
- PDF documentation

17. Limitations

The documented project uses the synthetic AI4I 2020 Predictive Maintenance Dataset rather than live industrial sensor streams. The AI functionality also depends on local Ollama model availability. Real industrial deployment would require additional validation, real-time sensor integration, production-grade identity management and operational security hardening.

18. Future Scope

- Real-time IoT sensor integration and live telemetry streaming.
- Dedicated machine-learning failure prediction models.
- Cloud synchronization and enterprise database integration.
- Email/SMS maintenance alerts.
- Advanced audit logging and granular permissions.
- Mobile-friendly technician interface.
- Predictive Remaining Useful Life (RUL) estimation.
- Integration with industrial maintenance / ERP systems.

19. Conclusion

FacilityOps AI Platform combines predictive-maintenance analytics, facility management workflows, AI assistance, authentication and role-based access control in one Streamlit application. The system connects **machine data** → **analysis** → **diagnosis** → **maintenance planning** → **work-order execution** → **documentation**.

The authentication evaluation procedure has been updated to remove the earlier one-click Demo Login instruction. The Administrator account is created first through the account-creation flow, after which both Administrator and Technician users access the application through the normal Sign In gateway.

20. Quick Reference for Evaluator

Launch: `streamlit run app.py`

Authentication: Create Administrator account first; use normal Sign In for Administrator and Technician.

Admin-only: EDA, Executive BI Dashboard, AI Maintenance Recommendations.

Shared operational modules: Home, Maintenance Status, Machine Explorer, Work Orders, Maintenance Schedules.

Core technologies: Streamlit, Python, Pandas, NumPy, Plotly, Ollama, ReportLab, SHA-256 and JSON.